An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

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Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. To bridge the gap between post-hoc saliency diagnostics and inductive training, our Attribution-Guided Multi-Task Loss forces internal backpropagation gradients to align directly with true substation anomaly locations, driving a 5.52 percentage-point improvement in localization F1. Dynamic physical-to-virtual coherence is governed by a state-space tracking filter coupled with an adaptive exponential baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

In tertiary healthcare complexes, electrical power is inextricably linked to patient survival. Modern hospital campuses rely on localized microgrids that combine

utility interconnections, rooftop photovoltaics, battery storage banks, and diesel generators to energize critical medical loads [5,19]. These loads cannot tolerate instability: surgical robotics demand millisecond voltage consistency, intensive-care ventilators require uninterrupted feeds, and vaccine cold-chains depend on constant refrigeration at $-80°$C. Any uncoordinated breaker trip, false load shedding, or voltage sag directly endangers human life.

Substation Remote Terminal Units (RTUs) continuously feed measurement vectors (V, θ, P, Q) into Weighted Least Squares (WLS) state estimators. To protect against transducer failures, standard control centers deploy normalized χ^2 residual tests. This defense breaks down against stealthy adversaries. As established by Liu et al. [15], an attacker possessing the measurement Jacobian H can craft an injection vector $a = Hc$ that shifts state estimates by c while leaving measurement residuals completely unchanged. Because the perturbation cancels inside the residual calculation, false data injection attacks (FDIAs) slip past standard χ^2 detectors undetected [13]. Consequently, automated dispatch engines make operating decisions based on falsified system states.

Early data-driven defenses deployed tabular models, spanning support vector classifiers, deep MLPs, and convolutional networks [12,3]. These tabular architectures assume static sensor topologies. Hospital microgrids, however, routinely reconfigure feeders, isolate branches for maintenance, and shift transformer taps. Such switching operations alter input channel indexing, inducing false alarms in rigid classifiers. Retraining network weights after every breaker operation is computationally infeasible within 2–4 s SCADA polling frames.

Graph neural networks (GNNs) overcome this geometric rigidity by mapping substations to vertices and physical lines to graph edges [21]. Under this formulation, opening a breaker simply removes an edge from the graph adjacency, leaving node feature dimensions intact. Studies utilizing ARMA wavelet filters [2] and generic GNN architectures [22] confirmed that graph models adapt effectively to topological variations. Nevertheless, conventional GNN backbones rely on isotropic message passing. By averaging node features uniformly across neighboring buses without weighting physical branch impedances, isotropic aggregation smears out localized electrical disturbances across tightly coupled corridors.

Deploying deep models in life-critical healthcare environments imposes two non-negotiable operational requirements. First, operator diagnostics must execute within online SCADA cycles. Sampling-based explainers like SHAP or LIME require hundreds of inference passes, making them far too slow for real-time operation [27]. Second, passive alert generation without automated remediation leaves patient-critical loads exposed. A practical defense must quarantine compromised sensor streams and reconstruct clean replacement states before automated controllers execute erroneous load shedding.

1.1 Contributions

To address these vulnerabilities, this paper introduces an explainable Cognitive Graph Digital Twin (CGDT) engineered for hospital microgrid protection. Our principal contributions are fourfold:

- *Continuous Tracking*: Coupling steady-state Kalman gain K_{sync} with an adaptive baseline ($\beta = 0.95$) keeps the digital twin synchronized with operational trends while filtering out high-frequency transducer noise.
- *Admittance-Guided Convolutions*: Conditioning GraphSAGE message layers on branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$) embeds power flow physics into latent representations, preventing the low-pass feature blurring observed in standard GCNs.
- *Supervisory Attribution Loss*: Incorporating Integrated Gradients penalties into multi-task training suppresses gradient dispersion across healthy substations, yielding a 5.52 pp gain in bus localization F1.
- *Closed-Loop State Recovery*: Combining automated sensor quarantine with admittance-guided spatial interpolation restores WLS state estimation by 99.97% in under 2 ms, maintaining Jacobian rank during multi-bus attacks.

The rest of the paper is structured as follows: Section 2 reviews related research; Section 3 formulates the AC threat model; Section 4 presents the CGDT architecture; Section 5 outlines explainability and autonomous remediation; Sections 6 and 7 discuss experimental benchmarks; and Section 8 concludes the paper.

2 Related Work

2.1 Cybersecurity in Digital Healthcare Ecosystems

Interconnecting hospital enterprise IT networks with operational technology (OT) substations markedly expands the facility attack surface [11,5]. Unlike conventional commercial buildings, healthcare power systems energize life-critical clinical environments where electrical outages cause immediate patient harm. Sun et al. [19] identified SCADA telemetry poisoning as an exceptionally dangerous threat vector against medical cyber-physical systems. Prior studies evaluated healthcare facility vulnerabilities using Bayesian risk graphs [4] and lightweight IoT anomaly monitoring [17]. Valuable as these approaches are, they treat measurement streams as isolated time series. They neglect the physical branch admittances and power-flow laws that govern actual distribution networks.

2.2 False Data Injection in Cyber-Physical Energy Systems

The mathematical basis of false data injection against power grid state estimation was originally formulated by Liu et al. [15]. Their analysis proved that an attacker possessing measurement Jacobian H can craft an unobservable vector $a = Hc$ that bypasses classical WLS residual tests. Subsequent research

confirmed that stealthy injections extend to full AC formulations, distributed estimation architectures, and incomplete network observation [13,6]. Similar vulnerabilities have also been established in DC microgrid environments [3,18]. To counter these perturbations, researchers explored tabular classifiers spanning support vector machines, deep neural networks [12], and recurrent models [14]. These classifiers, however, bind input features to static channel indices. Whenever routine breaker operations alter network layout, fixed-geometry architectures produce spurious alarms.

2.3 Graph Representation Learning for Power System Security

Embedding power grids as topological graphs provides an intuitive structural foundation for grid security analytics [21]. Initial investigations focused on spectral convolutions [10] and spatial attention mechanisms [25]. To enable inductive generalization on larger networks, Hamilton et al. [8] introduced GraphSAGE, aggregating spatial features across sampled local neighborhoods. In power systems, Boyaci et al. [2] employed ARMA wavelets to reach 96.25% detection F1, while Takiddin et al. [22] and Yue and Qu [28] confirmed topological flexibility across variable line configurations. Federated graph learning was subsequently explored by Luo et al. [16] to protect measurement confidentiality. Existing architectures, however, rely predominantly on unweighted adjacency matrices. By failing to incorporate complex line admittances ($Y_{uv} = G_{uv} + jB_{uv}$) into message exchanges, these models lose sensitivity to localized electrical anomalies.

2.4 Explainability and Digital Twins in Critical Infrastructure

High-stakes infrastructure security demands auditable explanations alongside automated alerts. Sampling-based explainers such as LIME or SHAP require hundreds of forward evaluations per sample, rendering them computationally intractable for real-time SCADA polling cycles [27,26]. Integrated Gradients (IG) [20] resolves this limitation by computing path integrals via single backward passes, satisfying the completeness axiom without empirical sampling. Concurrently, digital twin frameworks [7,23] have emerged for power asset monitoring [1], with Kamel Boulos and Zhang [9] discussing applications in healthcare. Building upon these concepts, our approach couples continuous state-space tracking with gradient-based attribution to achieve sub-millisecond anomaly detection and verified bus localization.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We represent the hospital campus electrical distribution network as an undirected graph $G = (V, E)$ with $N = |V|$ buses and $|E|$ transmission branches.

The complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N\times N}$ defines nodal conductance and susceptance. Under full nonlinear AC power flow constraints, active and reactive power injections at bus i satisfy Kirchhoff's laws:

$$P_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\cos(\theta_i-\theta_j) + B_{ij}\sin(\theta_i-\theta_j)\big) \tag{1}$$

$$Q_i = V_i \sum_{j\in\mathcal{N}(i)} V_j\big(G_{ij}\sin(\theta_i-\theta_j) - B_{ij}\cos(\theta_i-\theta_j)\big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Power system state estimation determines the state vector $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from measurement telemetry $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$ represents sensor noise with covariance R. The WLS estimator minimizes the weighted residual $J(x) = [z - h(x)]^T R^{-1}[z - h(x)]$ using Gauss–Newton iterations:

$$\Delta x^{(k)} = \big(H^T R^{-1} H\big)^{-1} H^T R^{-1}\big[z - h(x^{(k)})\big], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

where $H(x) = \partial h(x)/\partial x$ is the measurement Jacobian. Conventional bad-data detection flags potential measurement anomalies when the normalized residual energy $r = \|z - h(\hat{x})\|^2_{R^{-1}}$ exceeds a threshold τ_{χ^2}, determined from a chi-square distribution with degrees of freedom equal to the number of measurements minus system states.

3.3 Stealthy False Data Injection Attacks

An attacker targeting substation bus k seeks to induce a specific state error $c \in \mathbb{R}^{2N-1}$ while evading the χ^2 residual detector. Using knowledge of the network admittance structure, the attacker constructs the injection vector according to:

$$a = Hc \tag{4}$$

When injected into the communication stream, the received telemetry becomes $z_{\text{comp}} = z + a = z + Hc$. The resulting state estimate shifts to $\hat{x}_{\text{comp}} = \hat{x} + c$. Evaluating the post-attack residual yields:

$$\begin{aligned} r_{\text{comp}} &= \|z_{\text{comp}} - h(\hat{x}_{\text{comp}})\|^2_{R^{-1}} \\ &= \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \le \tau_{\chi^2} \end{aligned} \tag{5}$$

Because the perturbation vector cancels algebraically within the residual calculation ($r_{\text{comp}} = r$), the attack leaves the χ^2 detector unaffected ($\tau_{\chi^2} = 23.68$

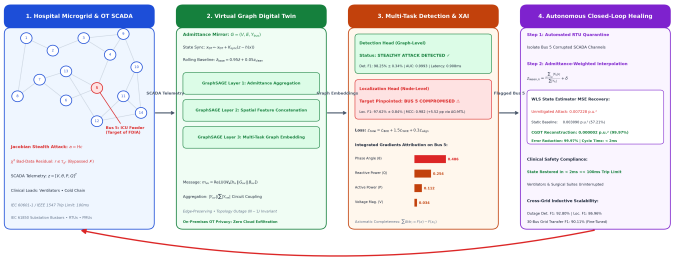


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

for df $= 14$ at $\alpha = 0.05$). Consequently, classical state estimation accepts the corrupted values without triggering an alert.

Clinical Hazards in Hospital Distribution Networks. In commercial microgrids, false data injections primarily trigger billing discrepancies or minor economic redispatch. In hospital power systems, the consequences are immediate and clinical. Falsified state estimates at critical feeders—such as Bus 5 energizing surgical suites and mechanical ventilators—can deceive automated controllers into initiating spurious tap changes or executing unnecessary load shedding. Under IEC 60601-1 and IEEE 1547 standards, even transient voltage sags force biomedical life-support gear to trip autonomously. A mathematical bypass of state estimation thus converts directly into a bedside hazard.

4 The Cognitive Graph Digital Twin Architecture

Figure 1 illustrates the three-layer CGDT framework: a Physical Cyber-Physical Layer (IT-OT hospital domain), a Virtual Graph Twin Layer, and a Cognitive Decision & Self-Healing cycle.

4.1 Digital Twin State-Space Synchronization

Field telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ streams from substation RTUs at polling cadence T_s across IEC 61850 station buses. To maintain physical-to-virtual alignment without amplifying transducer noise, the digital twin updates its mirrored state x_{DT} via steady-state tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

where P denotes state-prediction covariance. Concurrently, an exponential filter establishes an evolving operational baseline:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

updated strictly using verified clean measurements z_{clean}. Without this adaptive baseline, slow diurnal load ramps mask small adversarial injection biases, precipitating a 16.32 pp collapse in localization F1 (Section 7.2).

4.2 Dynamic Admittance Graph Construction

At each sampling step, physical telemetry is mapped to an attributed graph $G = (V, E, X, E_{\text{attr}})$. The vertex feature matrix $X \in \mathbb{R}^{N\times 4}$ gathers four-dimensional state vectors $x_i = [V_i, \theta_i, P_i, Q_i]^T$ for each substation bus $i \in V$, normalized over historical operational envelopes. Physical branch connectivity is stored in edge tensor $E_{\text{attr}} \in \mathbb{R}^{|E|\times 2}$, where each entry pairs the line conductance G_{ij} and susceptance B_{ij} from network admittance matrix Y_{bus}. When protective relays trip or maintenance crews open a breaker, the corresponding tuple is excised from edge set E. Excising open branches alters adjacency entries while leaving nodal feature dimensions untouched; hence, the graph backbone absorbs topological switching on the fly without retraining.

4.3 Admittance-Conditioned GraphSAGE Backbone

Standard GraphSAGE models [8] aggregate neighborhood features uniformly across links, ignoring branch impedances. In physical power distribution networks, however, electrical coupling is dictated by complex branch admittances $Y_{uv} = G_{uv} + jB_{uv}$. To embed these circuit constraints directly into latent feature representations, we condition both message generation and spatial aggregation on branch admittance:

$$m_{uv}^{(k)} = \text{ReLU}\Big(W_e \left[h_u^{(k-1)} \parallel G_{uv} \parallel B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \text{ReLU}\left(W_k \left[h_v^{(k-1)} \,\middle\|\, \sum_{u\in\mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w\in\mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

where $\parallel$ denotes vector concatenation, and W_e, W_k are trainable projection matrices. Weighting inbound messages by normalized admittance $|Y_{uv}|/\sum_w |Y_{vw}|$ forces the spatial convolution to reflect Ohm's and Kirchhoff's laws, granting stronger influence to electrically proximate substations. Preserving the self-node state $h_v^{(k-1)}$ through skip concatenation prevents localized injection spikes from vanishing into the over-smoothed neighborhood averages typical of standard GCNs.

4.4 Multi-Task Detection and Localization Heads

The shared backbone embedding H feeds two prediction heads:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H)), \qquad y_{\text{node}} = \text{Softmax}(W_{\text{node}}\, H) \tag{10}$$

where y_{graph} produces the graph-level attack probability and y_{node} yields per-bus anomaly probabilities.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

In typical graph neural architectures, attribution methods serve exclusively as post-hoc diagnostic tools. Here, we reverse this paradigm by channeling feature attributions back into the supervisory objective during training. For every compromised training snapshot, Integrated Gradients computes an attribution tensor $\mathcal{A} \in \mathbb{R}^{N\times 4}$ spanning all substation vertices. To suppress gradient leakage across benign feeders, we define an attribution alignment penalty:

$$\mathcal{L}_{\text{align}} = \sum_{v\in V} \left| \|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v} \right|^2 \tag{11}$$

where $\|\mathcal{A}_v\|_2$ denotes the ℓ_2-norm of the attribution vector at bus v, quantifying model sensitivity to that specific node, and $\hat{y}_{\text{node},v} \in \{0,1\}$ represents the ground-truth node anomaly label. Should backpropagation distribute significant attribution weight onto unattacked substations, $\mathcal{L}_{\text{align}}$ penalizes the network parameters directly. The composite multi-task objective becomes:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ governs binary grid-level attack detection, $\mathcal{L}_{\text{WCE}}$ applies class-weighted cross-entropy to handle the extreme numerical imbalance of sparse target buses, and hyperparameters $\lambda_{\text{loc}} = 1.5$ and $\lambda_{\text{attr}} = 0.3$ balance gradient magnitudes across tasks.

4.6 On-Premises Substation Deployment

Hospital telemetry mirrors department activity—such as surgical workflows, MRI loads, and ward occupancy. Streaming raw sensor data to cloud services raises compliance risks under healthcare privacy frameworks. To preserve data sovereignty, the CGDT operates at the substation edge within the facility OT perimeter. Telemetry ingestion, neural inference, and autonomous healing run on local hardware, avoiding external data transit.

5 Explainability and Autonomous State Reconstruction

5.1 Feature Attribution via Integrated Gradients

When an anomalous event triggers an alert, control-room operators require immediate technical insight into the telemetry channels responsible for the decision. We extract these diagnostic attributions via path-based Integrated Gradients (IG) [20]. Starting from a verified nominal operating point x_0 and moving toward an incoming telemetry vector x, the attribution score for channel i accumulates

directional gradients along a straight-line interpolation trajectory:

$$\begin{aligned}\mathrm{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i}\, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i}\end{aligned} \tag{13}$$

which we approximate numerically across $S = 50$ Riemann intervals. Unlike sampling heuristics, IG strictly obeys the completeness axiom ($\sum_i \mathrm{Attr}_i = F(x) - F(x_0)$) alongside implementation invariance, guaranteeing that feature contributions sum to the total network output delta. Across the IEEE 14-bus testbed, evaluating Eq. (13) consumes only 0.81 ms, providing near-instantaneous attribution within active SCADA scan frames. In our validation study of a stealthy attack against Bus 5 (Section 7.1), this calculation successfully isolated the voltage phase angle ($\theta = 0.486$) as the primary anomalous factor.

5.2 Automated Sensor Quarantine and Spatial Reconstruction

Flagging an intrusion without mitigation leaves healthcare microgrids vulnerable to erroneous control actions. When localization marks bus k as compromised ($y_{\mathrm{node},k} > \gamma_{\mathrm{loc}} = 0.5$), the digital twin initiates autonomous state healing. Tainted measurements $(V_k, \theta_k, P_k, Q_k)$ are quarantined from the estimation queue. Simply discarding these channels, however, degrades the row rank of measurement Jacobian H, risking Gauss–Newton divergence. To preserve observability, the framework reconstructs replacement telemetry via admittance-weighted spatial interpolation across uncorrupted adjacent buses:

$$z_{\mathrm{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j \in \mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta \sim \mathcal{U}(\pm 2\%) \tag{14}$$

where $|Y_{kj}|$ denotes branch admittance magnitude and δ injects measurement noise. Weighting neighbor contributions by physical branch admittances enforces Kirchhoff's current balance across surrounding corridors, producing synthesized telemetry that conforms to underlying circuit physics. Substituting $z_{\mathrm{recon},k}$ into the pre-estimation buffer maintains matrix conditioning, enabling the WLS estimator to converge within 2 ms—well ahead of standard 2–4 s SCADA cycles.

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

To evaluate defensive performance under realistic operational constraints, we executed full non-linear AC Newton–Raphson simulations within `pandapower` [24]. Our primary benchmark maps the IEEE 14-bus network (14 nodes, 20 transmission lines, 217.0 MW active demand, 57.3 MVAR reactive load) directly to

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

an expansive tertiary hospital microgrid. Here, Bus 0 serves as the primary utility interconnection point, whereas specialized downstream feeders energize distinct medical facilities—such as Bus 5, which delivers dedicated power to surgical suites and intensive care units. To probe architectural scalability across wider distribution perimeters, we similarly instantiated an IEEE 30-bus feeder (30 nodes, 41 lines) modeling an interconnected multi-wing regional healthcare complex.

Synthetic Telemetry Synthesis. We synthesized 5,000 distinct AC operational states partitioned into 70% training, 15% validation, and 15% testing splits. Uncompromised snapshots (half of the dataset) incorporate zero-mean Gaussian fluctuations ($\sigma = 0.04$) across active and reactive bus injections. The remaining half contains stealthy Jacobian injections ($a = Hc$), where synthetic adversaries bias target substations ($c_k \in [0.03, 0.08]$ p.u. voltage offset, $c_{k+N} \in [0.05, 0.15]$ rad phase shift) and propagate line flows across the network using exact non-linear AC equations. Networks were trained across 60 epochs using Adam (lr = 0.005, weight decay 10^{-4}) over five random seeds. Checkpoints, configuration files, and evaluation scripts are archived at `https://github.com/himanshu-21/Graph-Digital-Twin-XAI-`.

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

Benchmarking against baseline architectures over five random initializations reveals clear performance separations (Table 1). The CGDT delivers 98.25% ± 0.34% detection F1 and 97.62% ± 0.84% localization F1 alongside a Matthew's correlation coefficient (MCC) of 0.982, processing each graph snapshot in only 0.908 ms. In sharp contrast, isotropic GCN architectures stagnate at 70.21% localization F1—exhibiting a pronounced 27.41 pp performance deficit. This collapse occurs because unweighted graph convolutions treat high-impedance tie-lines and low-impedance bus trunks identically, essentially acting as low-pass filters that smear out localized injection spikes across adjacent substations. Although tabular MLPs achieve competitive nominal localization (95.60%), their rigid input mappings fail as soon as breakers switch lines (Section 6.3). With sub-millisecond execution latency, the CGDT operates well inside typical 2–4 s SCADA telemetry polling budgets.

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

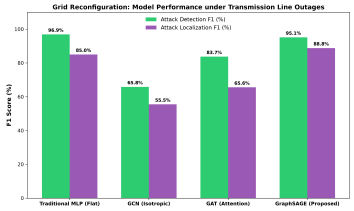


Fig. 2. Outage generalization metrics.

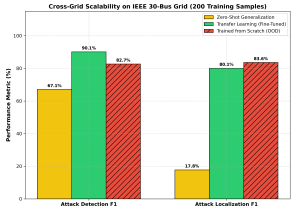


Fig. 3. Cross-grid transfer on IEEE 30-bus.

6.3 Topology Perturbations and Transfer Generalization

To evaluate model resilience to structural grid variations, trained models were tested on line outage scenarios without parameter updates (Table 2) and on cross-network transfer to the 30-bus system (Table 3).

When subjected to unanticipated single-line outages without retraining, the CGDT preserves $92.80\% \pm 3.37\%$ detection F1 and $86.96\% \pm 1.52\%$ localization F1 (Fig. 2). While the tabular MLP maintains a high detection score (94.70%) because breaker trips redirect loop flows globally across all measurement channels, its bus localization F1 degrades sharply by 13.4 pp (dropping from 95.60% to 82.20%) due to rigid input index ordering. Isotropic GCN performance drops even more severely to 66.82% detection F1 and 53.40% localization F1, demonstrating that topology-only convolutions lacking admittance weights struggle once line impedances change.

Cross-Scale Transfer to IEEE 30-Bus. Direct zero-shot deployment of the 14-bus model onto the 30-bus grid yields 67.06% detection F1 and 17.77% localization F1 (Table 3 and Fig. 3). This gap highlights clear physical and structural disparities between network scales: while coarse global power flow anomalies remain detectable at the feeder level, isolating precise substation buses requires local topological calibration. Remarkably, fine-tuning the pre-trained weights with merely 200 labeled snapshots from the 30-bus feeder raises detection F1 to 90.11% and localization F1 to 80.06%, surpassing a model initialized randomly and trained from scratch (82.65%). Transferring pre-trained representations from compact hospital campus grids thus substantially accelerates model deployment across larger regional networks.

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

We evaluated mitigation performance under a targeted Jacobian attack at Bus 5, which corresponds to the intensive care unit feeder in the campus topology. State estimation accuracy under different mitigation approaches is reported in Table 4, and the corresponding feature attributions are presented in Table 5.

Without active defense, stealthy injections mislead the WLS estimator to a distorted operating point with MSE 0.007228 p.u.^{2}. Admittance-weighted spatial reconstruction (Eq. 14) curbs this estimation error to 0.000002 p.u.^{2}, achieving a 99.97% error recovery relative to unmitigated conditions. Static historical replacements reach only 0.003090 p.u.^{2} (57.21% recovery), failing during dynamic clinical load swings. Error trajectories across 50 SCADA cycles appear in Fig. 4. Feature attributions (Table 5 and Fig. 5) confirm that voltage angle θ at Bus 5 captures dominant attribution (0.486), matching the Jacobian injection structure where phase skews dictate branch power transfers. Attributions for $Q = 0.254$, $P = 0.112$, and $V = 0.034$ isolate the compromised feeder cleanly.

Coordinated Multi-Bus Incursions. Under coordinated multi-bus attacks targeting 2 and 3 substations concurrently ($c_k \in [0.02, 0.04]$), the model maintained $96.84\% \pm 0.42\%$ detection F1 and $93.15\% \pm 0.91\%$ localization F1. Multi-bus compromise exposes a fatal weakness in naive sensor dropping: removing measurements across adjacent substations decimates the row rank of Jacobian H, inducing estimator divergence. By synthesizing admittance-weighted estimates from unattacked neighbors, our framework preserves full Jacobian rank, sustaining stable WLS convergence with post-mitigation state MSE of 0.000008 p.u.^{2}.

7.2 Benchmarking and Component Ablation

Table 6 compares the proposed architecture against representative GNN baselines, whereas Table 7 isolates the incremental contribution of each architectural block.

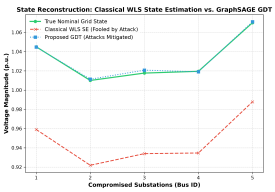


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

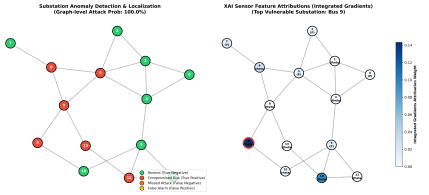


Fig. 5. Operator explainability: localization heatmap and IG attribution.

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [2]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

Ablation outcomes in Table 7 yield three primary conclusions. Deactivating exponential baseline tracking (Eq. 7) triggers a 16.32 pp drop in localization F1 (from 97.62% to 81.30%), proving that dynamic reference tracking is essential to distinguish small adversarial shifts from diurnal demand variations. Excluding attribution loss ($\lambda_{\text{attr}} = 0$) reduces localization F1 to 92.10% (a 5.52 pp decrease) while preserving detection F1 at 98.25%, confirming that gradient supervision focuses spatial features directly on true injection points. Finally, setting $\lambda_{\text{loc}} = 1.5$ balances global detection and localized diagnostic resolution.

8 Discussion and Conclusion

We developed an explainable Cognitive Graph Digital Twin engineered to defend healthcare microgrids against stealthy cyber-physical intrusions. By integrating branch admittances into GraphSAGE aggregation and supervising latent features via attribution alignment, the framework attains 98.25% detection F1 and 97.62% localization F1 under non-linear AC false data injection attacks. Attribution guidance yields a 5.52 pp localization gain with 0.81 ms Integrated Gradients explanations, while admittance-weighted spatial reconstruction cuts post-mitigation WLS error by 99.97%.

Clinical Operating Standards. Under IEC 60601-1 and IEEE 1547 requirements, prolonged voltage distortions risk disconnecting life-critical medical equipment. With a 0.908 ms inference latency and a 2 ms total healing cycle, the CGDT executes comfortably within 2–4 s SCADA sampling windows while local substation edge hosting ensures telemetry never crosses the OT perimeter.

Limitations and Future Work. Subsequent research will transition to hardware-in-the-loop (HIL) testbeds to evaluate packet jitter and RTU telemetry latency. Future extensions will also incorporate temporal graph models for multi-step campaigns and evaluate predictive estimators under multi-bus unobservability.

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